

Multiple Choice (10 marks)

1. Which of the following statements about lipids is correct?
 - (a) Lipids are the primary source of energy for cellular processes.
 - (b) Lipids are soluble in water but insoluble in organic solvents.
 - (c) Phospholipids are water soluble.
 - (d) Steroids are lipids that consist of glycerol and fatty acids.
 - (e) Lipids make up most cell surface receptors.
2. The theory of evolution is most accurately described as
 - (a) a universally accepted fact about the origin and development of all species.
 - (b) a hypothesis in the process of being tested and verified.
 - (c) a disproven theory about the development of species over time.
 - (d) a speculation about possible changes in populations over time.
 - (e) an opinion that some scientists hold about how living things change over time.
3. How can one demonstrate that amino acids are used to build proteins in cells, while the proteins already in the cell are used in catabolism?
 - (a) By observing the cell under a microscope
 - (b) Using fluorescent tagging of proteins
 - (c) Tracking changes in cell size and shape
 - (d) Autoradiography with labelled amino acids
 - (e) Through protein synthesis
4. Which of the following statements about mitochondria and chloroplasts is generally true?
 - (a) Plants, animals, and fungi all have both chloroplasts and mitochondria.
 - (b) Plants and fungi have chloroplasts but no mitochondria; animals have only mitochondria.
 - (c) Animals have both chloroplasts and mitochondria; plants and fungi have only chloroplasts.
 - (d) Plants and animals have chloroplasts but no mitochondria; fungi have only mitochondria.
 - (e) Plants and animals have mitochondria but no chloroplasts; fungi have only chloroplasts.

5. What are the typical periods in the unfolding of a behavioral act?
- (a) Appetitive behavior, exploratory behavior, quiescence
 - (b) Termination, appetitive behavior, exploratory behavior
 - (c) Appetitive behavior, consummatory act, quiescence
 - (d) Consummatory act, exploratory behavior, termination
 - (e) Appetitive behavior, termination, consummatory act

True / False (10 marks)

Answer True or False

- 6. True or False: Using negative reinforcement alone is sufficient to assess a dog's hearing sensitivity to various sound frequencies.
- 7. True or False: Down's syndrome can occur due to a mutation in chromosome 14, leading to altered genetic expression.
- 8. True or False: Tracheids and vessel elements are two xylem cell types that provide structural support and conduct water and minerals in plants.
- 9. True or False: Stabilizing selection promotes extreme traits, leading to increased variability in reproductive success among species.
- 10. True or False: The rise in intracellular free calcium in the sea urchin oocyte initiates the acrosomal reaction, allowing multiple sperm to fertilize the egg.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

- 11. Discuss the significance of courtship behaviors, such as the male common tern presenting a fish, in terms of communication and reproductive success in animal species.
- 12. Discuss the essential characteristics that differentiate living organisms from non-living matter, emphasizing the complexity and functionality required for life.

End of Paper